

Skills Summary

Inverse and Inverse-Trig Derivatives

Use this reference while completing your worksheet or when studying.

Skill 1 — The Three Inverse-Trig Rules

Rules (chain rule already built in, with u a function of x):

$$\frac{d}{dx} \arcsin u = \frac{u'}{\sqrt{1-u^2}}, \quad \frac{d}{dx} \arccos u = \frac{-u'}{\sqrt{1-u^2}}, \quad \frac{d}{dx} \arctan u = \frac{u'}{1+u^2}$$

Arcsine and arccosine differ only by a sign. Write down u and u' before you start, and remember to square u , not x , in the denominator.

Example: Differentiate $y = \arcsin(5x)$

Step 1. Name the inside function first: $u = 5x$, so $u' = 5$ — then the rule is a fill-in-the-blanks job.

$$\text{Step 2. } y' = \frac{5}{\sqrt{1-(5x)^2}} = \frac{5}{\sqrt{1-25x^2}}.$$

$$\Rightarrow y' = \frac{5}{\sqrt{1-25x^2}}$$

Try it: Differentiate $y = \arctan(4x)$

check: $\frac{4x}{1+16x^2}$

Skill 2 — The Inverse-Function Rule

Rule: if $f(a) = b$, then

$$(f^{-1})'(b) = \frac{1}{f'(a)}.$$

Order matters: find the input a that produces the output b , differentiate f , evaluate f' at a , then take the reciprocal. Evaluating at b is the standard error. You never need a formula for f^{-1} — which is what makes the rule worth having.

Example: $f(x) = x^3 + x + 6$ and $f(2) = 16$. Find $(f^{-1})'(16)$.

Step 1. The output is 16 and the input that gives it is 2, so the reciprocal must be taken of f' evaluated at 2, not at 16.

Step 2. $f'(x) = 3x^2 + 1$, so $f'(2) = 12 + 1 = 13$, and the inverse's slope is $\frac{1}{13}$.

$$\Rightarrow f'(x) = 3x^2 + 1 \quad \text{and} \quad (f^{-1})'(16) = \frac{1}{13}$$

Try it: $f(x) = x^3 + 2x + 5$ and $f(1) = 8$. Find $f'(x)$ and $(f^{-1})'(8)$.

check: $\frac{1}{13}$ and $3x^2 + 2$

Skill 3 — Inverse Trig Inside a Product

Rule: an inverse-trig function is just another factor, so the product, quotient and chain rules apply as usual. Expect the answer to keep an arcsin, arccos or arctan term — an answer with the inverse-trig function still in it is normal, not a sign that something went wrong.

Example: Differentiate $y = x \arcsin x$

Step 1. Two non-constant factors are multiplied, so this is the product rule with $u = x$ and $v = \arcsin x$.

Step 2. $y' = (1) \arcsin x + x \cdot \frac{1}{\sqrt{1-x^2}} = \arcsin x + \frac{x}{\sqrt{1-x^2}}.$

$$\Rightarrow y' = \arcsin x + \frac{x}{\sqrt{1-x^2}}$$

Try it: Differentiate $y = x \arccos x$

$\frac{d}{dx} x \arccos x = x \arccos x$ check